Yantian Zha

Assistant Professor, Department of Computer Science
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RESEARCH SPECIALIZATION

Embodied artificial intelligence; robotics; robot learning; machine learning; computer vision; planning; reinforcement learning; imitation learning; human-robot interaction; foundation models; robotic perception and control; simulation; and intelligent robotic systems operating across air, water, and land.

ACADEMIC APPOINTMENTS

Assistant Professor, Department of Computer Science **August 2025-Present**

North Carolina A&T State University

Tenure-track appointment. Director, Omnidomain AI and Robotics Laboratory.

Courses: COMP 645 Artificial Intelligence, Fall 2025 and Spring 2026.

Adjunct Lecturer **August 2023-Present**

University of Maryland, College Park

Courses taught include CMSC 421 Introduction to Artificial Intelligence; CMSC 848J Cognitive Robotics; and ENPM 808Z Cognitive Robotics.

Postdoctoral Associate **July 2024-August 2025**

University of Maryland, College Park

Supervisors: Professors Balakumar Balachandran and Miao Yu.

Postdoctoral Associate **July 2022-August 2023**

University of Maryland, College Park

Supervisor: Professor Yiannis Aloimonos.

Robotics Research Intern **March 2020-August 2020**

ABB Corporate Research Center, United States

Mentor: Dr. Jianjun Wang. Conducted robotics research and contributed software subsequently transferred to a product team.

Teaching Assistant **August 2016-December 2016**

Arizona State University

Course: Introduction to Artificial Intelligence.

EDUCATION

Ph.D. in Computer Science, Artificial Intelligence **2022**

Arizona State University, Tempe, Arizona

Inclusive dates: 2017-2022. Research advisor: Professor Subbarao Kambhampati.

M.S. in Computer Engineering, Computer Systems **2017**

Arizona State University, Tempe, Arizona

Inclusive dates: 2015-2017. Research advisor: Professor Subbarao Kambhampati.

B.E. in Electronics Engineering, Automation **2014**

Southeast University Chengxian College, China

Inclusive dates: 2010-2014. Research advisor: Professor Xudong Ma.

SPONSORED RESEARCH AWARDS AND SUPPORT

1. ERI: Towards Sustainable Aquaculture: Video-Based Reconstruction of Digital Underwater Animals for Robot Learning of Aquatic Capture Tasks. National Science Foundation, $200,000, July 1, 2026-June 30, 2028. Principal Investigator. Awarded in 2026.

2. Startup Package. North Carolina A&T State University, $110,000, July 1, 2026-June 30, 2028. Principal Investigator.

OTHER SPONSORED RESEARCH EXPERIENCE

• Planning-and-Plan-Recognition-Driven Attention Models for Supporting Active Perception in Decentralized Multi-Agent Systems. Office of Naval Research, 2018. Principal Investigator: Subbarao Kambhampati; Co-Principal Investigator: Baoxin Li. Contributing researcher.

PUBLICATIONS

Publications are grouped by publication type and listed in reverse chronological order within each category. An asterisk denotes equal contribution or an equal advising role, as indicated.

Peer-Reviewed Journal Articles

1. Zhuo, H. H., Zha, Y., and Kambhampati, S. (2019). Discovering Underlying Plans Based on Shallow Models. ACM Transactions on Intelligent Systems and Technology.

Peer-Reviewed Conference Papers

1. Shrestha, S., Zha, Y.*, Banagiri, S. N., Gao, G.*, Aloimonos, Y.*, and Fermuller, C.* (2025). NatSGLD: A Dataset with Speech, Gesture, Logic, and Demonstration for Robot Learning in Natural Human-Robot Interaction. ACM/IEEE International Conference on Human-Robot Interaction, pp. 1093-1098. *Authors indicated as equal research advisors in the supplied CV.

2. Guan, L., Zhou, Y., Liu, D., Zha, Y., Ben Amor, H., and Kambhampati, S. (2024). Task Success Is Not Enough: Investigating the Use of Video-Language Models as Behavior Critics for Catching Undesirable Agent Behaviors. Conference on Language Modeling. OpenReview.

3. Zha, Y.*, Guan, L.*, and Kambhampati, S. (2024). Learning from Ambiguous Demonstrations with Self-Explanation Guided Reinforcement Learning. Proceedings of the AAAI Conference on Artificial Intelligence, 38, 10395-10403. *Equal contribution.

4. Kambhampati, S., Sreedharan, S., Verma, M., Zha, Y., and Guan, L. (2022). Symbols as a Lingua Franca for Bridging Human-AI Chasm for Explainable and Advisable AI Systems. Proceedings of the AAAI Conference on Artificial Intelligence, Senior Member Presentation Track.

5. Zha, Y., Bhambri, S., and Guan, L. (2021). Contrastively Learning Visual Attention as Affordance Cues from Demonstrations for Robotic Grasping. IEEE/RSJ International Conference on Intelligent Robots and Systems, pp. 7835-7842.

6. Kulkarni, A., Zha, Y., Chakraborti, T., Vadlamudi, S. G., Zhang, Y., and Kambhampati, S. (2019). Explicability as Minimizing Distance from Expected Behavior. International Conference on Autonomous Agents and Multiagent Systems.

7. Zha, Y., Li, Y., Gopalakrishnan, S., Li, B., and Kambhampati, S. (2018). Recognizing Plans by Learning Embeddings from Observed Action Distributions. Proceedings of the 17th International Conference on Autonomous Agents and Multiagent Systems, pp. 2153-2155.

Workshop Papers

1. Zha, Y., Li, Y., Yu, T., Kambhampati, S., and Li, B. (2019). Plan-Recognition-Driven Attention Modeling for Visual Recognition. AAAI Workshop on Plan, Activity, and Intent Recognition.

Preprints

1. Yang, W.*, Kona, K.*, Jain, Y.*, Atzili, T.*, Bhamidipati, A., Lin, X., and Zha, Y. (2024). AAM-SEALS: Developing Aerial-Aquatic Manipulators in Sea, Air, and Land Simulator. arXiv preprint.

DATASETS AND SOFTWARE

• AAM-SEALS simulation project: aam-seals.umd.edu.

• NatSGLD dataset and project materials: snehesh.com/natsgld.

INVITED AND CONFERENCE PRESENTATIONS

1. Embodied and Human-Aligned Intelligence across Air, Water, and Land: From Simulation to Data-Efficient Learning. Spring Research Symposium for an NSF-supported Data Science and AI project, North Carolina A&T State University, May 16, 2025. Invited keynote presentation.

2. Cognitively-Enhanced Robotic Manipulation across Sea, Air, and Land. Maryland Research Day, College Park, Maryland, August 2024. Invited presentation.

3. Cognitively-Enhanced Robotic Manipulation across Sea, Air, and Land. Robotics Institute, Carnegie Mellon University, Pittsburgh, Pennsylvania, July 2024. Invited presentation.

4. Perceiving, Acting, Planning, and Self-Explaining: A Cognitive Quartet for Future Manufacturing. Maryland Robotics Center Research Symposium, College Park, Maryland, May 2024. Invited presentation.

5. Large Vision-Language Models for Self-Explanation-Guided Robot Learning. University of Maryland, College Park, Maryland, November 2023. Invited presentation.

6. Perceiving, Acting, Planning, and Self-Explaining: A Cognitive Quartet for Effective Human-Robot Collaboration. University of Central Florida, Orlando, Florida, March 2023. Invited presentation.

7. Contrastively Learning Visual Attention as Affordance Cues from Demonstrations for Robotic Grasping. IEEE/RSJ International Conference on Intelligent Robots and Systems, virtual, September 2021. Conference presentation.

8. Plan-Recognition-Driven Attention Modeling for Visual Recognition. AAAI Workshop on Plan, Activity, and Intent Recognition, Honolulu, Hawaii, January 2019. Conference presentation.

9. Recognizing Plans by Learning Embeddings from Observed Action Distributions. ICML/IJCAI/AAMAS Workshop on Planning and Learning, Stockholm, Sweden, July 2018. Conference presentation.

HONORS AND AWARDS

• Maryland Robotics Center Postdoctoral Fellowship, University of Maryland, 2022-2023. Project: Self-Explanation-Guided Deep Reinforcement of Imitation Learning from Human Advice for Task-Driven Perception and Robotic Manipulation.

PROFESSIONAL AND UNIVERSITY SERVICE

Peer Review and Professional Service

• Reviewed two conference papers in 2025. Additional reviewer service has included IEEE Robotics and Automation Letters; Conference on Robot Learning; Robotics: Science and Systems; International Conference on Learning Representations; International Conference on Automated Planning and Scheduling; and IEEE Transactions on Image Processing.

• Workshop Organizer, Bridging the Gap between Cognitive Science and Robot Learning in the Real World: Progresses and New Directions, Conference on Robot Learning, November 2023.

• Invited to serve as an Advisory Board member for a proposed human-computer interaction project at North Carolina A&T State University, 2026. Advisory participation is contingent on project funding.

• Undergraduate Research Thesis Discussant, Gemstone Honors Program, University of Maryland, March 2024.

University and Outreach Service

• Member, AI Working Group, North Carolina A&T State University, 2025-Present. Contributed to development of the university AI strategy supporting the 2030 mission.

• Laboratory tours and robotics research demonstrations for visitors.

STUDENT MENTORING AND ADVISING

North Carolina A&T State University

• Hangong Chen: Ph.D. student, March 2025-Present.

• Lee Caesar: Ph.D. student, August 2025-Present.

• Tahsin Zarman: Ph.D. student, January 2026-Present.

University of Maryland, College Park

• Yangsheng: Ph.D. student, co-mentored, August 2024-Present.